

Xingfan Xia

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WORK EXPERIENCE

Compute Labs, Remote, Redmond, WA

- Co-Founder and Chief Technology Officer Aug 2024 – Present
 - **Leading the technological vision for democratizing AI compute access through blockchain-based tokenization, building the financial ecosystem for compute as an emerging asset class.**
 - Architecting the Compute Tokenization Protocol (CTP) that transforms physical GPUs (H100s, B200s) into tradeable digital assets, enabling fractional ownership and enhanced liquidity in AI infrastructure investments.
 - Directing R&D teams in developing the AI-Fi ecosystem that merges advanced AI infrastructure with DeFi protocols, creating innovative financial instruments and structured products for AI compute investments.
 - Spearheading the development of GNFT & COMPUTE token systems, implementing multi-faceted validation and benchmarking frameworks to ensure transparent, verifiable, and secure GPU tokenization transactions.
 - Building scalable blockchain infrastructure to optimize GPU utilization across decentralized and centralized compute networks, maximizing returns for investors throughout the AGI development journey.

Amazon Web Service – Athena, Redmond, WA

- Software Engineer - AI/ML Infrastructure Apr 2022 – Jun 2024
 - **Building next-generation data infrastructure optimized for AI/ML workloads, enabling large-scale machine learning analytics on petabyte-scale datasets.**
 - **Tech Stack: Java, Python, Apache Spark, Apache Iceberg, AWS ML Services, Distributed Computing**
 - Architected CTAS feature for Iceberg Tables enabling efficient ML data preprocessing. Released at AWS RE:INVENT 2022 with millions of ML-focused queries executed since launch.
 - Optimized Hive JSON Serde library for ML data ingestion, improving feature extraction performance by 30% and reducing ML pipeline failures by 60%.
 - Developed virtualized table connector for ML feature stores, improving AI application data accessibility by 40% and projected to save \$4M annually in ML infrastructure costs.

Apple, Cupertino, CA

- Machine Learning Engineer - Siri Intelligence Aug 2020 – Apr 2022
 - **Developing advanced AI systems and machine learning models to enhance Siri's intelligence across iOS, macOS, watchOS, and tvOS platforms.**
 - **Tech Stack: Python, Objective-C, TensorFlow, PyTorch, MLOps, Apache Spark, Real-time ML Inference**
 - Designed on-device ML framework for user profiling using deep learning algorithms, serving as core intelligence layer for Apple Maps and Apple News recommendation systems.
 - Built automated MLOps pipeline for continuous model training with A/B testing framework and automated performance monitoring, reducing manual ML engineering overhead by 40%.
 - Developed contextual AI prediction service using ensemble learning to anticipate user intent, powering Siri Shortcuts and proactive recommendations.

Airbnb, San Francisco, CA

- Machine Learning Engineer - Trust & Safety AI Jan 2019 – Aug 2020
 - **Building AI-powered fraud detection and trust & safety systems using advanced machine learning techniques and real-time inference engines.**
 - **Tech Stack: Python, Java, Scala, TensorFlow, XGBoost, MLflow, Apache Kafka, Real-time ML Serving**
 - Developed ML algorithms for fake review detection using NLP and anomaly detection, reducing fraudulent reviews to 1% of peak volume and preventing \$2M in losses.
 - Implemented ensemble learning for comprehensive user risk scoring system, combining multiple ML models to evaluate host and guest trustworthiness.
 - Deployed ML-driven policy engine for automated risk-based payout delays, preventing 60% of fraudulent payouts and generating \$10M in annual savings.
 - Architected real-time ML serving infrastructure providing low-latency risk signals through Kafka streams and Redis caching for sub-100ms inference.

EDUCATION

Carleton College, Northfield, Minnesota

- B.A. in Computer Science Graduated Jun 2018

TECHNICAL SKILLS

- Programming Languages: Proficiency in Python, Java, Golang, Javascript, Scala, Objective-C
- Experienced AWS Services: Athena, S3, SQS, EC2, Lambda, API Gateway, Auto Scaling Group, Cloudfront, Route 53, CodePipeline, CodeBuild, CodeDeploy, Kinesis
- Data Engineering & Analysis: Data Analysis Libraries, OpenCV, Visualization Tools, Scipy, Numpy, Pandas, ELK, Apache Iceberg, Apache Spark, Apache Kafka, Airflow

